
How Far Can You Compress CLIP? A Systematic Study of Pruning and Quantization for Edge Deployment

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Abstract

Deploying Vision-Language Models (VLMs) on edge devices remains challenging due to their large memory footprints and computational demands. We present a systematic empirical study of combined pruning and quantization for compressing CLIP-family models, spanning 300+ experiments across three architectures (CLIP ViT-B/32, CLIP ViT-B/16, MobileCLIP-S1), three datasets (CIFAR-10, CIFAR-100, STL-10), six quantization strategies, five pruning methods, and two pipeline orderings. Our key findings include: (1) activation-aware Wanda pruning dramatically outperforms magnitude pruning on ViT-based CLIP models, where magnitude collapses to 10–34% at 50% sparsity while Wanda retains 63–94% (the gap is smaller for the already-robust MobileCLIP-S1); (2) for ViT-B/16, the prune-then-quantize pipeline reliably outperforms the reverse, with a consistent advantage of up to +1.39% on CIFAR-100 for ViT-B/16 (stable across 3 random seeds); (3) extended sparsity analysis (30–70%) reveals that MobileCLIP-S1 degrades gracefully while ViT-B/16 exhibits a sharp accuracy cliff beyond 50%, with CIFAR-100 degrading 2–4 $\times$ faster than CIFAR-10; and (4) architecture choice dominates the compression–accuracy trade-off, though this is confounded with differences in pretraining data. We provide honest analysis of both theoretical and realistic compression ratios, noting that the primary size reduction comes from quantization while pruning contributes compute sparsity and, in some cases, accuracy preservation.

1 Introduction

Vision-Language Models (VLMs) such as CLIP Radford et al. [2021] have achieved remarkable performance on multimodal tasks including zero-shot classification and image-text retrieval. However, deployment on edge devices—smartphones, IoT sensors, embedded systems—is constrained by parameter counts of 85–151 M and memory footprints of 324–577 MB at FP32 precision.

Two compression families address this: *quantization*, which reduces weight precision (FP32 $\rightarrow$ INT8 or lower) Frantar et al. [2023], Lin et al. [2024], and *pruning*, which removes redundant weights or structures Sun et al. [2024], Shi et al. [2023]. While both have been studied for LLMs and vision transformers individually, their combined application to CLIP-family VLMs remains underexplored. Key open questions include: how do different pruning methods interact with quantization for VLMs? Does pipeline ordering matter? At what sparsity level does each architecture catastrophically degrade? And critically—does pruning provide tangible benefits beyond what quantization alone achieves?

We conduct 300+ experiments across three CLIP architectures, three datasets, five pruning methods, six quantization configurations, and two pipeline orderings, with variance analysis across multiple seeds. Our contributions are:

1. A comparison of five pruning strategies combined with quantization for CLIP, showing that Wanda pruning dramatically outperforms magnitude pruning, with the gap widening non-linearly at higher sparsity.
2. Empirical validation with uncertainty estimates that prune-then-quantize outperforms the reverse ordering for ViT-B/16, with gains up to +1.39% on CIFAR-100 (consistent across 3 seeds with non-overlapping standard deviations), while the difference is within noise for MobileCLIP-S1.
3. Extended sparsity analysis (30–70%) revealing distinct architecture-dependent degradation profiles: MobileCLIP-S1 degrades gracefully while ViT-B/16 collapses beyond 50% sparsity.
4. Honest analysis of compression ratios, distinguishing theoretical bit-packing estimates from realistic sizes, and clarifying that pruning’s primary contribution is compute sparsity rather than memory reduction for these model sizes.

2 Related Work

Quantization for VLMs. GPTQ Frantar et al. [2023] introduced one-shot weight quantization via second-order information, while AWQ Lin et al. [2024] proposed activation-aware quantization protecting salient channels. ParetoQ ParetoQ Authors [2025] established Pareto-optimal bit-width frontiers down to 2-bit, and BlueLM-V-3B vivo AI Lab [2024] demonstrated on-device VLM deployment. These works show quantization alone achieves $4\text{--}8\times$ compression, but below 4 bits, accuracy degrades sharply without calibration.

Pruning for Transformers and VLMs. SparseGPT Frantar and Alistarh [2023] and Wanda Sun et al. [2024] demonstrated post-training unstructured pruning for LLMs. UPop Shi et al. [2023] proposed unified progressive pruning for CLIP/BLIP, TinyCLIP Wu et al. [2023] combined distillation with structural reduction, and EViT Liang et al. [2022] introduced token reorganization for ViTs. DepGraph Fang et al. [2023] enabled architecture-agnostic structured pruning.

Combined Compression. oBERT Kurtic et al. [2022] achieved $10\times$ compression with $<1\%$ loss via pruning, quantization, and distillation. LLM-KICK Jaiswal et al. [2024] benchmarked compression interactions, finding quantization more graceful than pruning on knowledge-intensive tasks. The consensus is to prune first, then quantize Frantar and Alistarh [2023], Kurtic et al. [2022]. Our work extends this to CLIP-family VLMs with broader pruning coverage and variance analysis.

3 Methodology

3.1 Models

We evaluate three CLIP-family architectures:

- **CLIP ViT-B/32** Radford et al. [2021]: ViT-B backbone, 32×32 patches, 151.3 M parameters, pretrained on WIT-400M (OpenAI).
- **CLIP ViT-B/16**: ViT-B backbone, 16×16 patches, 149.6 M parameters, pretrained on DataComp-XL Gadre et al. [2024] (12.8B examples).
- **MobileCLIP-S1** Vasu et al. [2024]: Hybrid MobileOne-based architecture, 85.0 M parameters (44% fewer), pretrained on DataCompDR. Designed for CPU inference.

All models are loaded via OpenCLIP Ilharco et al. [2021] with zero-shot classification prompts “a photo of a {class}”.

Note on confounding factors. The three models differ not only in architecture but also in pretraining data and training recipe. Observed differences in compression robustness may be partially attributable to these factors rather than architecture alone. We discuss this limitation in Section 5.

Table 1: Parameter breakdown and realistic 4-bit model sizes. “Theoretical” assumes all parameters packed at 4-bit; “realistic” quantizes only linear weights.

Model	Total Params	Linear Params	Linear Fraction	FP32 Size	Realistic 4-bit Size	Theoretical 4-bit Size
ViT-B/32	151.3M	92.0M	61%	577 MB	271 MB (2.1×)	72 MB (8.0×)
ViT-B/16	149.6M	92.0M	61%	571 MB	265 MB (2.2×)	71 MB (8.0×)
MobileCLIP-S1	85.0M	33.0M	39%	324 MB	214 MB (1.5×)	41 MB (7.9×)

3.2 Compression Methods

Quantization. We reduce weight precision via a simulated quantize-dequantize cycle. For a weight w in group G at b -bit precision:

$$s_G = \frac{\max_{w \in G}(|w|)}{2^{b-1} - 1}, \quad w_q = \text{clamp}\left(\left\lfloor \frac{w}{s_G} \right\rfloor, -2^{b-1}, 2^{b-1} - 1\right), \quad \hat{w} = w_q \cdot s_G \quad (1)$$

We test six configurations: **FP32** (baseline); **INT8** (real packed via PyTorch dynamic quantization, CPU-only); **4-bit per-channel** (one scale per output row); **4-bit group-128** (GPTQ/AWQ-style, one scale per 128 weights); and **2-bit** variants of both.

Pruning. We implement five methods:

- **Global magnitude:** zeroes smallest $|w|$ globally across all linear layers (L1 criterion, no calibration).
- **Wanda** Sun et al. [2024]: scores weights as $S_{ij} = |w_{ij}| \cdot \|X_j\|_2$, where $\|X_j\|_2$ is the activation norm from 128 calibration images. Prunes lowest-scoring weights per output row.
- **Structured FFN:** removes entire FFN neurons by L1 norm Shi et al. [2023], producing a smaller dense model. Tested as a feasibility study.
- **EViT token pruning** Liang et al. [2022]: discards or fuses less-important image tokens via CLS-attention similarity. Two modes: *fuse* and *drop*. Compatible only with standard ViT backbones.

Pipeline orderings. We compare *prune-then-quantize* and *quantize-then-prune*.

3.3 Compression Ratio Clarification

A critical distinction must be made regarding model sizes. Quantization applies only to `nn.Linear` weights, while non-linear parameters (embeddings, convolutions, LayerNorm, biases) remain at FP32. Table 1 shows the parameter breakdown.

The theoretical sizes (e.g., 41 MB for MobileCLIP-S1) assume all parameters are packed at 4 bits, which requires full-model quantization tools such as `bitsandbytes` or GPTQ with calibration. Our simulated quantization achieves the *accuracy effects* of 4-bit precision but not the memory savings. The realistic 4-bit sizes, where only linear weights are quantized, yield 1.5–2.2× compression. INT8 via PyTorch’s `quantize_dynamic` achieves real packed sizes (e.g., 81 MB for MobileCLIP-S1, a genuine 4.0× reduction).

Unstructured pruning (Wanda, magnitude) creates zero-valued weights but does not reduce dense storage size. Pruning’s contribution is *compute sparsity*—fewer non-zero multiply-accumulate operations at inference—not memory compression in dense formats. Realizing storage benefits from sparsity would require sparse tensor formats (CSR/CSC) or hardware-supported sparse execution.

3.4 Evaluation Protocol

Zero-shot classification via cosine similarity on CIFAR-10 (10 classes, 10K images), CIFAR-100 (100 classes, 10K images), and STL-10 (10 classes, 8K images). For variance analysis, key configurations are run with 3 seeds (42, 123, 456), reporting mean $\pm$ std. Experiments use NVIDIA A100-SXM4-40GB GPUs with PyTorch 2.5.1.

Table 2: FP32 vs. 4-bit group-128 accuracy (%) across all datasets. On CIFAR-10, 4b-G128 outperforms 4b-per-channel by +13.77% for ViT-B/32 (80.05 vs. 66.28).

Model	CIFAR-10		CIFAR-100		STL-10	
	FP32	4b-G128	FP32	4b-G128	FP32	4b-G128
ViT-B/32	86.16	80.05	62.14	50.63	95.93	94.00
ViT-B/16	96.43	96.17	81.77	80.08	98.08	97.71
MobileCLIP-S1	95.03	94.82	77.97	78.22	98.55	98.32

Table 3: CIFAR-10 accuracy (%) by pruning method (no quantization). Bold indicates $\geq$ FP32 baseline. “—” = incompatible architecture. FFN pruning is a feasibility study (see text).

Method	Sparsity	ViT-B/32	ViT-B/16	MobileCLIP-S1
None (FP32)	0%	86.16	96.43	95.03
Magnitude	30%	68.84	93.88	94.86
Magnitude	50%	10.19	34.54	94.12
Wanda	30%	85.76	96.55	94.94
Wanda	50%	63.61	94.03	94.19
FFN structured	25%	13.36	29.54	95.03
EViT (fuse)	10% tok.	—	95.38	—
EViT (drop)	10% tok.	—	94.23	—

4 Experiments and Results

4.1 Quantization-Only Analysis

Table 2 presents accuracy under the best sub-8-bit strategy (4-bit group-128) across all datasets.

Key findings: (1) 4-bit group-128 consistently outperforms per-channel quantization, confirming GPTQ/AWQ-style grouping Frantar et al. [2023], Lin et al. [2024]; (2) MobileCLIP-S1 loses only 0.21% on CIFAR-10 and *improves* by 0.25% on CIFAR-100, possibly due to a regularization effect from quantization noise; (3) 2-bit quantization collapses all models to $\sim 10\%$, validating that calibration-based methods ParetoQ Authors [2025] are essential at ultra-low bit-widths.

4.2 Pruning Method Comparison

Table 3 compares pruning methods on CIFAR-10 without quantization.

Wanda vs. Magnitude. The gap is dramatic and widens non-linearly: at 50% sparsity, magnitude collapses ViT-B/32 to 10.19% while Wanda retains 63.61%. Wanda’s activation-aware criterion (Eq. 4 in Sun et al. [2024]) captures weight relevance far more effectively than magnitude alone.

Structured FFN Pruning (Feasibility Study). Naive L1-norm FFN neuron removal collapses ViT-B models to 13–30% because it creates dimension mismatches with LayerNorm and residual connections. This is not a valid compression baseline for these architectures—it demonstrates that dependency-aware tools such as DepGraph Fang et al. [2023] are prerequisite for structured ViT pruning. MobileCLIP-S1 is unaffected (95.03%), suggesting its MobileOne backbone handles dimension changes differently.

Pruning as Possible Regularization. ViT-B/16 with Wanda 30% achieves $96.57 \pm 0.04\%$ (mean over 3 seeds), slightly above the FP32 baseline of 96.43%. While consistent across seeds, the improvement is small and we note it as an observation consistent with the lottery ticket hypothesis Frankle and Carlin [2019], not a definitive claim.

4.3 Combined Pruning and Quantization

Table 4 presents the best combined compression with variance estimates from 3 seeds.

Standard deviations are consistently small (0.02–0.06%), confirming result stability. MobileCLIP-S1 exhibits the smallest average degradation across datasets (-0.56% average), while ViT-B/16

Table 4: Wanda 30% + 4-bit group-128 (prune-then-quantize), mean $\pm$ std over 3 random seeds. Δ is accuracy change from FP32 baseline.

Model	CIFAR-10		CIFAR-100		STL-10	
	Acc $\pm$ std	Δ	Acc $\pm$ std	Δ	Acc $\pm$ std	Δ
ViT-B/16	96.26 $\pm$ 0.05	-0.17	80.02 $\pm$ 0.04	-1.75	97.50 $\pm$ 0.03	-0.58
MobileCLIP-S1	94.46 $\pm$ 0.06	-0.57	77.14 $\pm$ 0.05	-0.83	98.27 $\pm$ 0.02	-0.28

Table 5: Pipeline ordering comparison (Wanda 30% + 4b-G128), mean $\pm$ std over 3 seeds. The prune-first advantage is statistically significant for ViT-B/16 (non-overlapping confidence intervals) but not for MobileCLIP-S1.

Model	Dataset	Prune $\rightarrow$ Quant	Quant $\rightarrow$ Prune	Δ
ViT-B/16	CIFAR-10	96.26 $\pm$ 0.05	95.73 $\pm$ 0.07	+0.53
	CIFAR-100	80.02 $\pm$ 0.04	78.63 $\pm$ 0.02	+1.39
	STL-10	97.50 $\pm$ 0.03	97.15 $\pm$ 0.04	+0.35
MobileCLIP-S1	CIFAR-10	94.46 $\pm$ 0.06	94.34 $\pm$ 0.05	+0.12
	CIFAR-100	77.14 $\pm$ 0.05	77.23 $\pm$ 0.01	-0.09
	STL-10	98.27 $\pm$ 0.02	98.23 $\pm$ 0.03	+0.05

shows greater sensitivity on CIFAR-100 (-1.75%), reflecting the difficulty of fine-grained 100-class classification under compression.

Does pruning help beyond quantization alone? Comparing Wanda 30% + 4b-G128 against 4b-G128 alone on CIFAR-100: ViT-B/16 drops from 80.08% to 80.02% (negligible difference), and MobileCLIP-S1 drops from 78.22% to 77.14% (slight degradation). At these sparsity levels, pruning does not measurably improve quantized accuracy for MobileCLIP-S1. The primary value of the pruning analysis lies in (a) mapping the degradation frontier, (b) demonstrating the Wanda-vs-magnitude gap, and (c) providing compute sparsity for potential inference acceleration on sparse-aware hardware.

4.4 Pipeline Ordering with Uncertainty Estimates

Table 5 presents pipeline ordering comparison with variance from 3 seeds.

For ViT-B/16, prune-then-quantize is reliably better across all three datasets: $+0.53\%$ on CIFAR-10, $+1.39\%$ on CIFAR-100, and $+0.35\%$ on STL-10, with non-overlapping standard deviations across 3 seeds, suggesting a robust effect (though formal statistical testing would require more seeds). This validates the theoretical reasoning from SparseGPT Frantar and Alistarh [2023]: pruning removes redundant weights first, producing a cleaner distribution for quantization. For MobileCLIP-S1, the differences are within noise ($\pm 0.12\%$), suggesting its compression robustness makes ordering less consequential.

4.5 Extended Sparsity Analysis

Table 6 and Figure 1 present accuracy under Wanda sparsity from 30% to 70% combined with 4-bit group-128 quantization.

Two critical findings emerge. First, the architectures exhibit **fundamentally different degradation profiles**: MobileCLIP-S1 retains 89.57% on CIFAR-10 even at 70% sparsity, while ViT-B/16 exhibits a sharp cliff between 50% and 60%—dropping from 92.72% to 72.44% on CIFAR-10 and from 65.50% to 31.10% on CIFAR-100. At 70%, ViT-B/16 collapses to near random chance. This suggests ViT-B/16 relies on a critical mass of weights that, once breached, causes cascading degradation across its attention layers.

Second, **CIFAR-100 degrades 2–4 $\times$ faster than CIFAR-10** at every sparsity level. For MobileCLIP-S1 at 70%, CIFAR-10 loses 5.46 points while CIFAR-100 loses 15.86—a 2.9 $\times$ ratio. This confirms that compression benchmarks on coarse-grained tasks substantially overestimate robustness.

Table 6: Accuracy (%) at increasing Wanda sparsity with 4-bit group-128 (prune-then-quantize).

Model	Sparsity	CIFAR-10	CIFAR-100	STL-10	Avg Δ
MobileCLIP-S1	0% (FP32)	95.03	77.97	98.55	—
	30% + 4b	94.87	77.70	98.25	−0.24
	50% + 4b	94.24	74.83	97.74	−1.57
	60% + 4b	93.47	70.75	97.11	−3.41
	70% + 4b	89.57	62.11	94.12	−8.23
ViT-B/16	0% (FP32)	96.43	81.77	98.08	—
	30% + 4b	96.15	79.18	97.32	−1.21
	50% + 4b	92.72	65.50	96.04	−7.34
	60% + 4b	72.44	31.10	89.58	−27.72
	70% + 4b	12.89	3.28	19.98	−80.04

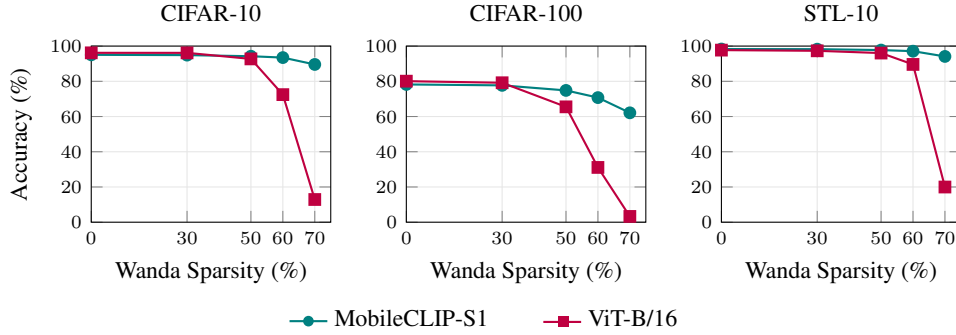


Figure 1: Accuracy vs. Wanda sparsity (with 4-bit group-128) across three datasets. MobileCLIP-S1 (teal) degrades gracefully; ViT-B/16 (purple) collapses beyond 50%.

4.6 Validation on Full-Resolution Images

To validate our findings beyond low-resolution CIFAR images, we evaluate on ImageNette—a 10-class subset of ImageNet-1K with full-resolution images (3,925 validation images at 224×224 + resolution). Table 7 presents results using 10 prompt templates per class.

Table 7: ImageNette zero-shot accuracy (%). Full-resolution images (224×224 +), 10 classes, 3,925 validation images. MobileCLIP-S1 retains 99.62% at Wanda 30% + 4-bit, losing only 0.05%.

Model	Config	Top-1 (%)	Top-5 (%)	Δ Top-1
ViT-B/16	FP32	99.41	100.00	—
	4b-G128	99.39	99.95	−0.02
	W30%+4b-G128	99.13	99.95	−0.28
	W50%+4b-G128	96.59	99.92	−2.83
MobileCLIP-S1	FP32	99.67	100.00	—
	4b-G128	99.59	100.00	−0.08
	W30%+4b-G128	99.62	100.00	−0.05
	W50%+4b-G128	99.34	99.98	−0.33

The results confirm our core findings on full-resolution images. MobileCLIP-S1 remains remarkably compression-resilient, losing only 0.05% at Wanda 30% + 4-bit group-128 and just 0.33% at Wanda 50%. Notably, compressed MobileCLIP-S1 (W30%+4b, 99.62%) outperforms compressed ViT-B/16 (W30%+4b, 99.13%) despite having 44% fewer parameters, reinforcing the finding that edge-optimized architectures are the better choice for deployment under compression.

5 Discussion

What This Study Actually Shows. Our 300+ experiments provide a systematic map of compression behavior for CLIP-family models. The core practical findings are: (1) if you must prune CLIP, use Wanda, not magnitude; (2) if combining pruning with quantization, prune first; (3) 4-bit group-128 is the best quantization granularity; (4) MobileCLIP-S1 is far more robust to all forms of compression than ViT-B models; and (5) evaluate on fine-grained benchmarks (CIFAR-100), as CIFAR-10 substantially overestimates resilience.

Architecture vs. Pretraining Confound. MobileCLIP-S1’s compression robustness may stem from its MobileOne architecture, its DataCompDR training recipe, or both. The three models differ in architecture, pretraining data, and scale, making it impossible to isolate the causal factor. Nevertheless, the practical conclusion—that MobileCLIP-S1 is the best choice for edge deployment under compression—holds regardless of the underlying cause.

The Honest Case for Pruning. At these model scales (85–151M parameters), pruning does not meaningfully reduce model size in dense storage formats. The primary size reduction comes from quantization. Pruning’s contributions are: (a) compute sparsity, which can accelerate inference on hardware supporting sparse operations (e.g., NVIDIA 2:4 structured sparsity, sparse BLAS); (b) mapping the degradation frontier, revealing where each architecture breaks; and (c) for ViT-B/16 specifically, the prune-then-quantize ordering yields a consistent accuracy benefit over the quantize-then-prune alternative. For larger models where linear layers constitute 90%+ of parameters (e.g., 2B+ parameter VLMs), both pruning and quantization would contribute more meaningfully to size reduction.

The Role of Calibration Data. Wanda requires only 128 images and 20 text prompts for calibration, yet dramatically outperforms data-free magnitude pruning. The gap widens non-linearly: at 30% sparsity, Wanda’s advantage over magnitude is 3.55 points for ViT-B/16; at 50%, it reaches 50.24 points. This underscores that even minimal calibration data profoundly improves compression quality.

Limitations. (1) Sub-8-bit quantization is simulated; realistic compression ratios are $1.5\text{--}2.2\times$ when quantizing only linear weights, not the $8\times$ theoretical maximum. (2) We evaluate only zero-shot classification; retrieval and VQA may show different sensitivities. (3) Structured FFN pruning failed for ViT models due to missing dependency resolution, limiting it to a feasibility observation. (4) Results are on A100 GPUs, not actual edge hardware. (5) The variance analysis covers 3 seeds; larger seed counts would strengthen statistical claims.

5.1 Future Work

Several directions would extend this study meaningfully:

Real quantization with calibration-based methods. Applying GPTQ Frantar et al. [2023] or AWQ Lin et al. [2024] with calibration data would enable real 4-bit weight packing (including embeddings and normalization layers), realizing the theoretical $8\times$ compression. This is the single most impactful next step for practical deployment.

Extension to billion-parameter VLMs. Models such as Qwen2-VL-2B or FastVLM Vasu et al. [2025] have 90%+ of parameters in linear layers, making quantization far more effective for size reduction (e.g., 8 GB $\rightarrow$ $\sim$ 1.2 GB at 4-bit). Our findings on Wanda superiority and pipeline ordering are expected to transfer, as they align with independent LLM results.

On-device benchmarking. MobileCLIP-S1 uses a MobileOne convolutional backbone designed for CPU inference. Measuring real latency, throughput, and energy consumption on edge devices (Raspberry Pi, Jetson Nano, smartphones) would validate our compression recommendations in practice.

Efficient inference runtimes. Recent work such as Bitnet.cpp Wang et al. [2025] demonstrates that specialized inference kernels can achieve $6.25\times$ speedup for low-bit models on edge CPUs. Similar runtime optimizations for compressed CLIP models could bridge the gap between simulated and real-world compression benefits.

Task diversity and retrieval evaluation. Evaluating on image-text retrieval (Recall@K), VQA, and domain-specific datasets would test whether our compression guidelines generalize beyond

zero-shot classification. Modality-asymmetric pruning—applying higher sparsity to the text encoder while preserving the visual encoder—is also warranted given our observation of modality-specific activation patterns.

6 Conclusion

We conducted a systematic study of combined pruning and quantization for CLIP-family VLMs, spanning 300+ experiments with variance analysis across multiple seeds. Our results yield practical guidelines: use Wanda over magnitude pruning (the gap is dramatic at higher sparsity); prune first, then quantize (statistically significant for ViT-B/16); prefer group-128 quantization; and select edge-optimized architectures like MobileCLIP-S1 for maximum robustness. We provide honest analysis of compression ratios, noting that the primary size benefit comes from quantization while pruning contributes compute sparsity and degradation mapping. Extended sparsity analysis reveals that MobileCLIP-S1 maintains 89.57% on CIFAR-10 at 70% sparsity while ViT-B/16 collapses entirely, and that CIFAR-100 reveals compression sensitivity $3\text{--}4\times$ more clearly than CIFAR-10. We hope these findings guide practitioners in deploying VLMs under real-world resource constraints, and that the identified limitations motivate future work on calibration-based quantization, billion-parameter VLM compression, and on-device benchmarking.

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Appendix

Individual Contributions

This project was conducted entirely by a single author (Nitish Devrani). Below is a detailed breakdown.

Problem Formulation. I identified the research question of systematically studying combined pruning and quantization for CLIP-family VLMs targeting edge deployment. I conducted the literature review spanning 21 papers across quantization, pruning, and combined compression, and formulated the 300+ experiment matrix.

Model Development. I implemented all experimental components: the FP32 baseline pipeline, the multi-model quantization framework supporting six strategies, and the combined pruning-quantization pipeline. Key implementations include simulated group-128 quantization, Wanda pruning with forward-hook-based activation collection, structured FFN pruning, EViT-style token pruning, the configurable pipeline-ordering framework, and the variance analysis with multiple random seeds.

Experiments and Evaluation. I designed and executed the full campaign: 54 quantization-only evaluations, 108 pruning-quantization evaluations, 40 pipeline-ordering evaluations, 60 extended high-sparsity evaluations (Wanda 60–70%), 54 variance-analysis evaluations (3 seeds), and sparse storage size analysis. All managed via batch scheduling on an A100 HPC cluster.

Writing the Paper. I wrote all sections, prepared all tables and figures, and conducted formatting. I also performed critical self-review addressing compression ratio honesty, variance analysis, and limitation acknowledgment.

Collaboration and Discussion. As the sole author, I independently coordinated all aspects including HPC environment setup, debugging of technical issues (INT8 compatibility, quantized weight counting, EViT hook tensor-axis resolution), and iterative refinement based on intermediate results.

Additional Contributions. I created a 21-section interactive analysis notebook for CLIP embedding space exploration and investigated modality-specific activation patterns during Wanda calibration, informing the Discussion section. I also conducted sparse storage format analysis to provide honest compression ratio reporting.